

TRIPLE QUANTUM SCHUBERT CALCULUS

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ABSTRACT.

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1. INTRODUCTION

2. DEFINITIONS AND STATEMENT OF RESULTS

Definition 2.1 ($\xrightarrow{k}$, $P_k(u, w)$, $q_k(u, w)$, $n_k(u, w)$). We define a relation $\xrightarrow{k}$ on pairs of permutations, introduced in its non-quantum form by Sottile [2].

Given $u, u' \in S_\infty$ and a positive integer k , we declare that $u \xrightarrow{k} w$ if there exists a p with $0 \leq p \leq k$ such that there are transpositions $t_{a_1 b_1}, \dots, t_{a_p b_p}$ satisfying

- (1) $a_i \leq k < b_i$ for all i
- (2) $a_i \neq a_j$ whenever $i \neq j$
- (3) $b_i \geq b_j$ whenever $i < j$
- (4) We have either

$$\ell(ut_{a_1 b_1} \cdots t_{a_i b_i}) = \ell(ut_{a_1 b_1} \cdots t_{a_{i-1} b_{i-1}}) + 1$$

or

$$\ell(ut_{a_1 b_1} \cdots t_{a_i b_i}) = \ell(ut_{a_1 b_1} \cdots t_{a_{i-1} b_{i-1}}) - 2(b_i - a_i) + 1$$

for all i

- (5) $w = ut_{a_1 b_1} \cdots t_{a_p b_p}$

If $u \xrightarrow{k} w$, define

$$P_k(u, w) = \{u(i) \mid i \leq k \text{ and } u(i) = w(i)\}$$

Also, we define $q_k(u, w)$ recursively. If $u = w$, define $q_k(u, w) = 1$. If we have defined $q_k(u, w')$ with $w' = t_{a_1 b_1} \cdots t_{a_{p-1} b_{p-1}}$, suppose $\ell(w' t_{a_p b_p}) = \ell(w') + 1$. Then define

$$q_k(u, w) = q_k(u, w')$$

If instead $\ell(w' t_{a_p b_p}) = \ell(w') - 2(b_p - a_p) + 1$, define

$$q_k(u, w) = q_k(u, w') q_{a_p b_p}$$

We also define

$$n_k(u, w) = \#\{i \mid i \leq k \text{ and } u(i) \neq w(i)\}$$

Theorem 2.1 (The Pieri formula). *Suppose $u \in S_\infty$ and $p \leq k$. Then*

$$\mathfrak{S}_u^q(x; y) E_{p, k}^q(x; z) = \sum_{u \xrightarrow{k} w} q_k(u, w) E_{p - n_k(u, w), k - n_k(u, w)}(y_{P_k(u, w)}; z) \mathfrak{S}_w^q(x; y)$$

Theorem 2.2. *Suppose $u, v \in S_\infty$ and let d be a weak composition such that $d_i \in \{0, 1\}$ for all i and if $d_i = 1$, then $d_{i+1} = 0$. Let $i_1, \dots, i_k$ be the integers such that $d_{i_p} = 1$ for $1 \leq p \leq k$. Then $c_{u, v}^{w, d}(y; z) = 0$ unless $\ell(us_{i_p}) < \ell(u)$ and $\ell(vs_{i_p}) < \ell(v)$ for each p , in which case*

$$c_{u, v}^{w, d}(y; z) = c_{us', vs'}^w(y; z)$$

where

$$s' = s_{i_1} \cdots s_{i_k}$$

Definition 2.2 ($\mathbf{c}$, (strict) dominant permutations, θ , μ , $\tilde{\theta}$, $\tilde{\mu}$). We say that a permutation μ is *dominant* if $\mathbf{c}_i(\mu) \geq \mathbf{c}_{i+1}(\mu)$ for all i . A permutation μ is said to be *strict dominant* if μ is dominant and whenever $\mathbf{c}_i(\mu) \neq 0$ we have $\mathbf{c}_i(\mu) > \mathbf{c}_{i+1}(\mu)$. For a permutation $v \in S_\infty$, we define a dominant permutation $\mu(v)$ recursively as follows. If v is already dominant, define $\mu(v) = v$. Otherwise, let i be the maximal index such that $\mathbf{c}_i(v) < \mathbf{c}_{i+1}(v)$, and define

$$\mu(v) = \mu(vs_i)$$

Define $\theta_i(v)$ to be $\mathbf{c}_i(\mu(v))$.

Similarly, for a permutation $v \in S_\infty$, we define a strict dominant permutation $\tilde{\mu}(v)$ recursively as follows. If v is already strict dominant, define $\tilde{\mu}(v) = v$. Otherwise, let i be the maximal index such that $\mathbf{c}_{i+1}(v) \neq 0$ and $\mathbf{c}_i(v) \leq \mathbf{c}_{i+1}(v)$, then define

$$\tilde{\mu}(v) = \tilde{\mu}(vs_i)$$

Define $\tilde{\theta}_i(v)$ to be $\mathbf{c}_i(\tilde{\mu}(v))$.

Definition 2.3 ($\overset{i}{\Rightarrow}$, $\sigma_i(u, v)$, $D_i(u, v)$). Let $u, v \in S_\infty$ and $i \geq 1$ be an integer. We define $u \overset{i}{\Rightarrow} v$ if there exist distinct integers $b_1, b_2, \dots, b_k$ such that $b_j \neq i$ for all j , $\ell(ut_{i,b_1} \cdots t_{i,b_k}) = \ell(u) + k$ for all j , and

$$v = ut_{i,b_1} \cdots t_{i,b_k}$$

We define

$$\sigma_i(u, v) = \#\{j \mid b_j < i\}$$

and we define

$$D_i(u, v) = \{u(i)\} \cup \{u(j) \mid u(j) \neq v(j)\}$$

Theorem 2.3 (Formula for ordinary double Schubert polynomials). *Suppose $v \in S_\infty$. Then*

$$\mathfrak{S}_v(x; z) = \sum_{(v_0, \dots, v_k)} \prod_{i=1}^k (-1)^{\sigma_i(v_{i-1}, v_i)} E_{\theta_i(v) - \ell(v_{i-1}, v_i), \theta_i(v)}(x; z_{D_i(v_{i-1}, v_i)})$$

Where $v_{i-1} \overset{i}{\Rightarrow} v_i$ for all i , $v_0 = 1$, and $v_k = v\mu(v^{-1})$.

Theorem 2.4 (Formula for quantum double Schubert polynomials). *Suppose $v \in S_\infty$. Then*

$$\mathfrak{S}_v^q(x; z) = \sum_{(v_0, \dots, v_k)} \prod_{i=1}^k (-1)^{\sigma_i(v_{i-1}, v_i)} E_{\tilde{\theta}_i(v) - \ell(v_{i-1}, v_i), \tilde{\theta}_i(v)}^q(x; z_{D_i(v_{i-1}, v_i)})$$

Where $v_{i-1} \overset{i}{\Rightarrow} v_i$ for all i , $v_0 = 1$, and $v_k = v\tilde{\mu}(v^{-1})$.

Coupled with the Pieri formulas, these formulas give a relatively efficient method for computing products of (quantum) double Schubert polynomials.

Definition 2.4 ($\chi_{j;n}$, $\chi^{(n)}$). Recall the definition of the commuting difference operators $\chi_{j;n}$, with $j \leq n$, defined by Fomin, Gelfand, and Postnikov, as follows:

$$\chi_{j,n} = x_j - \sum_{1 \leq i < j} q_{ij} \partial^{t_{ij}} + \sum_{j < k \leq n} q_{jk} \partial^{t_{jk}}$$

We denote the sequence $(\chi_{1;n}, \dots, \chi_{n;n})$ with n fixed by $\chi^{(n)}$. Recall (CITATION) that if $\mathfrak{S}_v(x)$ has at most n variables, then

$$\mathfrak{S}_v^q(\chi^{(n)})(1) = \mathfrak{S}_v(x)$$

and hence

$$\mathfrak{S}_v^q(\chi^{(n)}; z)(1) = \mathfrak{S}_v(x; z)$$

We specify a more precise formula, as follows.

Theorem 2.5. *Suppose $v \in S_n$. Then*

$$\mathfrak{S}_v^q(\chi^{(n)}; z) = \sum_{u \in S_n, d} q^d c_{u,v}^{1,d}(x; z) \partial^u$$

Proof. Note that if $\mathfrak{S}_u(x)$ has at most n variables, then

$$\mathfrak{S}_u^q(\chi^{(n)}; y) \mathfrak{S}_v^q(\chi^{(n)}; z) = \sum_{w,d} q^d c_{u,v}^{w,d}(y; z) \mathfrak{S}_w^q(\chi^{(n)}; y)$$

We thus have that

$$(\mathfrak{S}_u^q(\chi^{(n)}; y) \mathfrak{S}_v^q(\chi^{(n)}; z))(1) = \sum_{w,d} q^d c_{u,v}^{w,d}(y; z) \mathfrak{S}_w(x; y)$$

This is also equal to

$$\mathfrak{S}_v^q(\chi^{(n)}; z)(\mathfrak{S}_u(x; y))$$

due to the fact that $(\mathfrak{S}_u^q(\chi^{(n)}; y))(1) = \mathfrak{S}_u(x; y)$. To obtain the coefficient of ∂^u in $\mathfrak{S}_v^q(\chi^{(n)}; z)$, we set $y = x$ after applying this operator. Since $\mathfrak{S}_w(x; x) = 0$ unless $w = 1$, we have that

$$\mathfrak{S}_v^q(\chi^{(n)}; z)(\mathfrak{S}_u(x; y))|_{y=x} = \sum_d q^d c_{u,v}^{1,d}(x; z)$$

and the result follows. $\square$

Theorem 2.6. *Let k be an integer and let $v, w \in S_\infty$. Then*

$$\sum_d q^d c_{s_k, v}^{w,d}(y; z) = c_{s_k, v}^w(y; z) + \sum_{i \leq k < j} q_{ij} \mathfrak{S}_{vt_{ij}w^{-1}}(y; z)$$

Proof. We have that the desired value is equal to the coefficient of $\mathfrak{S}_w(x; y)$ in

$$\mathfrak{S}_{s_k}^q(\chi^{(n)}; y)(\mathfrak{S}_v(x; z))$$

We have that

$$\mathfrak{S}_{s_k}^q(\chi^{(n)}; y) = \mathfrak{S}_{s_k}(x; y) + \sum_{i \leq k < j \leq n} q_{ij} \partial^{t_{ij}}$$

Applying this operator, we obtain

$$\sum_{i \leq k < j \leq n} q_{ij} \mathfrak{S}_{vt_{ij}}(x; z) + \sum_w c_{s_k, v}^w(y; z) \mathfrak{S}_w(x; y)$$

Applying the Cauchy formula, this is equal to

$$\sum_{i \leq k < j, u} q_{ij} \mathfrak{S}_{vt_{ij}u^{-1}}(y; z) \mathfrak{S}_u(x; y) + \sum_w c_{s_k, v}^w(y; z) \mathfrak{S}_w(x; y)$$

Setting $u = w$ and allowing n to be sufficiently large, we obtain the result. $\square$

Proof of Theorem 2.2. It suffices to prove this for $v = w_0(n)$. We have

$$\mathfrak{S}_{w_0(n)}^q(\chi; z) = \sum_{u,d} q^d c_{u, w_0(n)}^{1,d}(x; z) \partial^u$$

Fixing the particular d in the statement of the theorem, we must have that $u = s'$. By the Pieri formula,

$$c_{s', w_0(n)}^{1,d}(x; z) = \mathfrak{S}_{w_0(n)s'}(x; z)$$

Thus if we compute the coefficient of $q^d \mathfrak{S}_w(x; y)$ in

$$(\mathfrak{S}_{w_0(n)}^q(\chi; z))(\mathfrak{S}_u(x; y))$$

we obtain

$$c_{us', w_0(n)s'}^w(y; z)$$

and the result follows. $\square$

3. PROOFS

Lemma 3.1. Suppose $(v_0, v_1, \dots, v_n)$ and $(v'_0, \dots, v'_n)$ are such that $v_{i-1} \xrightarrow{i} v_i$, $v'_{i-1} \xrightarrow{i} v'_i$ for all i , $v_0 = v'_0 = 1$, and $v_n = v'_n = v$ for some n . Suppose also that $D_i(v_{i-1}, v_i) = D_i(v'_{i-1}, v'_i)$ for all i . Then $v_i = v'_i$ for all i .

Proof. This is true if $\ell(v_n) = 0$. Otherwise, consider the right descent. There are only two places where it can occur. $\square$

Theorem 3.2. No two terms in Theorem 2.3 directly cancel each other.

Proof. $\square$

4. PROOF OF THE PIERI FORMULA

Lemma 4.1. Let $u \in S_\infty$ and let $i, j \geq 1$. Then

$$\begin{aligned} (x_i - z_j) \mathfrak{S}_u^q(x; y) &= (y_{u(i)} - z_j) \mathfrak{S}_u^q(x; y) + \sum_{\substack{p \neq i \\ \ell(ut_{ip}) = \ell(u) + 1}} \text{sign}(p - i) \mathfrak{S}_{ut_{ip}}^q(x; y) \\ &+ \sum_{\substack{p \neq i \\ \ell(ut_{ip}) = \ell(u) - 2|p - i| + 1}} \text{sign}(p - i) q_{ip} \mathfrak{S}_{ut_{ip}}^q(x; y) \end{aligned}$$

Definition 4.1 (Descents of a cycle, $\text{desc}_u(c)$). Let c be a cycle and suppose $u \in S_\infty$. Write c as

$$(a_1, \dots, a_n)$$

where $a_n > a_i$ for all $i < n$ (this is a unique way to express the cycle since the a_j are distinct). Then for any i with $1 \leq i < n$ we say that a_i is a *descent* of c with respect to u if $u(a_i) > u(a_{i+1})$. We denote the set of all descents of c with respect to u by $\text{desc}_u(c)$.

Lemma 4.2. Suppose $u, w \in S_\infty$ and let $k > 0$ be an integer. Then $u \xrightarrow{k} w$ if and only if there exist pairwise disjoint cyclic permutations $c_1, \dots, c_n$ of respective cycle lengths $p_1 + 1, \dots, p_n + 1$ such that

$$w = uc_1 \cdots c_n$$

and

$$\ell(u, w) = p_1 + \cdots + p_n$$

such that for each i there exist pairwise distinct positive integers $a_{i,1}, \dots, a_{i,p_i}$ and b_i such that we may write the cycle c_i as

$$c_i = (a_{i,p_i}, a_{i,p_i-1}, \dots, a_{i,1}, b_i)$$

where $a_{i,j} \leq k$ for all $1 \leq j \leq p_i$, $b_i > k$, and $b_i > b_j$ if $i < j$, and for each m we have

$$\ell(uc_1 \cdots c_m) = \sum_{i=1}^m \left(p_i - 2 \sum_{a \in \text{desc}_u(c_i)} (b_i - a) \right)$$

Proof. Suppose first that $u \xrightarrow{k} w$. Then there exist reflections $t_{a_1 b_1}, \dots, t_{a_m b_m}$ for some m with $a_i \neq a_j$ if $i \neq j$, $a_i \leq k < b_i$ for all i , $\ell(ut_{a_1 b_1} \cdots t_{a_i b_i}) = \ell(t_{a_1 b_1} \cdots t_{a_{i-1} b_{i-1}}) + 1$ or $\ell(ut_{a_1 b_1} \cdots t_{a_i b_i}) = \ell(t_{a_1 b_1} \cdots t_{a_{i-1} b_{i-1}}) - 2(b_i - a_i) + 1$ for all i , $b_i \geq b_j$ if $i \leq j$, and $w = ut_{a_1 b_1} \cdots t_{a_m b_m}$. Fix b such that there is an index $1 \leq p \leq m$ such that $b = b_p$ and let i and r be such that $a_i, \dots, a_{i+r}$ is the maximal portion of the sequence such that $b_j = b$ for all $i \leq j \leq i+r$. Let

$$c = t_{a_i b_i} \cdots t_{a_{i+r} b_{i+r}}$$

Given j with $i \leq j \leq i+r$ we claim that if we define a permutation $c^{(j)}$ with

$$c^{(j)} = t_{a_i b} \cdots t_{a_j b}$$

then

$$c^{(j)} = (a_j, a_{j-1}, \dots, a_i, b)$$

and that

$$\ell(uc^{(j)}) = \ell(u) + j - 2 \sum_{a \in \text{desc}_u(c^{(j)})} (b - a)$$

This is clear if $j = i$. Otherwise, suppose

$$c^{(j-1)} = t_{a_i b} \cdots t_{a_{j-1} b} = (a_{j-1}, \dots, a_i, b)$$

Then

$$c^{(j-1)} t_{a_j b} = (a_{j-1}, \dots, a_i, b)(b, a_j) = (a_{j-1}, \dots, a_i, b, a_j) = (a_j, \dots, a_i, b)$$

as desired. We have either that

$$u(a_j) = uc^{(j-1)}(a_j) < uc^{(j-1)}(b) = u(a_{j-1})$$

and $\ell(uc^{(j)}) = \ell(uc^{(j-1)}) + 1$, or

$$u(a_j) > u(a_{j-1})$$

and $a_j \in \text{desc}_u(c^{(j)})$ and $\ell(uc^{(j)}) = \ell(uc^{(j-1)}) - 2(a_j - b) + 1$. Hence the desired claim follows by induction. Hence,

$$c = c^{(i+r)} = (a_{i+r}, \dots, a_i, b)$$

and the length satisfies the condition. Ranging over all b , these cycles, which we index in order of decreasing b as $c_1, \dots, c_n$, are the ones that occur in the disjoint cycle decomposition of $u^{-1}w$, which we can see by the fact that the cycles are pairwise disjoint and $w = uc_1 \cdots c_n$.

We prove the converse now. Suppose the cycles $c_1, \dots, c_n$ of cycle length $p_i + 1$ for each i exist satisfying the conditions, fixing k . Write c_1 as

$$c_1 = (a_{p_1}, \dots, a_1, b)$$

Then

$$c_1 = t_{a_1 b} \cdots t_{a_{p_1} b}$$

We claim that $\ell(ut_{a_1 b} \cdots t_{a_i b}) > \ell(ut_{a_1 b} \cdots t_{a_{i-1} b})$ if $a_i \notin \text{desc}_u(c_1)$, and $\ell(ut_{a_1 b} \cdots t_{a_i b}) < \ell(ut_{a_1 b} \cdots t_{a_{i-1} b})$ if $a_i \in \text{desc}_u(c_1)$. This clearly holds for a_1 . Otherwise, we note that

$$ut_{a_1 b} \cdots t_{a_{i-1} b}(b) = u(a_{i-1})$$

and

$$ut_{a_1 b} \cdots t_{a_{i-1} b}(a_i) = u(a_i)$$

since $a_j \neq a_i$ if $j \neq i$. Thus, if $a_i \notin \text{desc}_u(c_1)$, then the length increases, and if $a_i \in \text{desc}_u(c_1)$ then the length decreases. Applying this argument for each disjoint cycle c_i to $\ell(uc_1 \cdots c_i)$ and expanding the cycles into reflections, we obtain that successively multiplying on the right by each reflection increases the length by at least 1 for each a_i that is not a descent and decreases the length by at most $2(b_i - a) - 1$ for each descent in c_i . By the length condition in the statement of the lemma, the increases must be by exactly 1 and the decreases by exactly $2(b_i - a) - 1$ by maximality. $\square$

Definition 4.2 ($\triangleleft$). If $u, u' \in S_\infty$, we define $u \triangleleft u'$ if $\ell(u, u') = 1$ or $u' = ut_{ab}$ where $\ell(u, u') = 1 - 2(b - a)$.

Lemma 4.3. Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer. Suppose $u \xrightarrow{k} w$ and let $c = (a_p, \dots, a_1, b)$ be a cycle as in Lemma 4.2 in the decomposition of $u^{-1}w$. Then $\text{desc}_u(c)$ has at most one element.

Proof. This is clearly true if $p = 1$, or if the result holds for $p - 1$ but $c^{(p-1)}$ has no descents. Suppose then that $c^{(p-1)}$ has a descent, say a_i for exactly one $i < p$ by the induction hypothesis, and that $u(a_p) > u(a_{p-1})$, meaning that $a_p \in \text{desc}_u(c^{(p-1)})$. a_i being a descent would require that

$$u(a_i) > uc^{(i-1)}(j) > u(a_{i-1})$$

for all j with $a_i < j < b$. Furthermore, we also have that for all j with $a_p < j < b$ that

$$u(a_p) > uc^{(p-1)}(j) > u(a_{p-1})$$

Suppose $a_p < a_i$. Then $u(a_p) > uc^{(p-1)}(a_i) = u(a_{i-1}) > u(a_{p-1})$. This means that necessarily $a_{p-1} < a_i$, since $u(a_{i-1}) = uc^{(i)}(a_i)$ is minimal in the range $a_i \leq j \leq b$ in $uc^{(i)}$. For the same reason, we cannot have that $uc^{(p-2)}(b) < u(a_{i-1})$, because this would have to come from some $t_{a_q b}$ with $i < q < p - 1$, $a_q < a_i$, and $u(a_q) < u(a_{i-1})$, which would increase the length by more than 1. However, since

$uc^{(p-2)}(b) > u(a_{i-1}) > u(a_{p-1})$, this would imply that $t_{a_{p-1}b}$ increases the length by more than 1, which is a contradiction. Thus we cannot have a descent a_p with $a_p < a_i$.

If instead $a_p > a_i$, then $u(a_i) > u(a_p) > u(a_{i-1})$. Since a_p is a descent, $uc^{(p-1)}(b) = u(a_{p-1}) < u(a_p)$, but $uc^{(i)}(b) = u(a_i) > u(a_p)$. This would imply that there is some q with $i < q \leq p-1$ such that $a_q > a_p$ and $uc^{(q-1)}t_{a_qb}(b) = u(a_q) < u(a_p)$, because if we had $a_q < a_p$ then the length would increase by more than 1. Let q be minimal with this property. In that case, $u(a_{q-1}) > u(a_p)$ and $uc^{(p-1)}(a_q) = u(a_{q-1})$, contradicting the fact that a_p is a descent. It follows by induction that there can be at most one descent. $\square$

Definition 4.3 (Redefinition of $\text{desc}_u(c)$). If c is a cycle, we redefine $\text{desc}_u(c)$ to be the unique element of the previously defined set $\text{desc}_u(c)$ if it exists, or otherwise 0.

Lemma 4.4. Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer. Suppose $u \xrightarrow{k} w$ and let $c = (a_p, \dots, a_1, b)$ be a cycle as in Lemma 4.2 in the decomposition of $u^{-1}w$, with $a_q = k$ being a descent. Then for all $j \geq q$ we have that $u(a_j) > u(b)$.

Proof. We have that $u(a_q) > u(a_{q-1})$. By Lemma 4.3, k is the only descent of c . We claim that for all r with $k \leq r < b$ we have that $u(r) > u(b)$. This is because for all such r we have that $u(a_{q-1}) < u(r)$ since a_q is a descent, and if $u(b) > u(r)$ for some such r then we may pick a maximal $\ell < q$ such that $u(a_\ell) > u(r)$ for some such r . Then $u(a_{\ell+1}) < u(r)$ for all such r , and the reflection $t_{a_{\ell-1}a_\ell}$ would increase the length by more than 1, which cannot happen. For the same reason, we cannot have that $u(a_m) < u(r)$ for any r with $k < r \leq b$, since $u(a_q) > u(r)$ for all such r and hence the reflection $t_{a_{m-1}a_m}$ would increase the length by more than q . Thus since in particular $u(r) > u(b)$ for all such r , we must have that $u(a_j) > u(b)$ for all $j \geq q$, as desired. $\square$

Lemma 4.5. Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer. If $u \xrightarrow{k} w$ but $u \not\xrightarrow{k-1} w$, then there is a unique $b > k$ such that $wt_{kb} \triangleleft w$ and $u \xrightarrow{k-1} wt_{kb}$. Furthermore, in that case there does not exist a $a < k$ such that $wt_{ak} \triangleleft w$ and $u \xrightarrow{k-1} wt_{ak}$.

Proof. Suppose $u \xrightarrow{k} w$ but $u \not\xrightarrow{k-1} w$. Let $c_1, \dots, c_n$ be the pairwise disjoint cycles in the decomposition of $u^{-1}w$, with the cycle length of c_i being $p_i + 1$. Some such cycle must contain k , as otherwise we would have $u \xrightarrow{k-1} w$ by Lemma 4.2. Assume that $c_i = (a_{i,p_1}, \dots, a_{i,1}, b_i)$ is the cycle that contains k and set $b = b_i$. We claim that $u \xrightarrow{k-1} wt_{kb}$ and $wt_{kb} \triangleleft w$. Note that $w = (wt_{kb})t_{kb}$ and hence

$$(uc_1 \cdots c_n t_{kb})t_{kb} = w$$

Let j be the index such that $a_j = k$. Then we have

$$\begin{aligned} (a_{i,p_i}, \dots, a_{i,j+1}, k, a_{i,j-1}, \dots, a_{i,1}, b_i)t_{kb} &= (a_{i,j-1}, \dots, a_{i,1}, b, a_{i,p_i}, \dots, a_{i,j+1}, k)t_{kb} \\ &= (a_{i,j-1}, \dots, a_{i,1}, b)(a_{i,p_i}, \dots, a_{i,j+1}, k) \end{aligned}$$

Note that $wt_{kb}(k) = w(b) = u(a_{i,p_i})$ and $wt_{kb}(b) = w(k) = u(a_{i,j-1})$. We need that

$$\ell(u, wt_{kb}) = p'_i + p''_i - 2(b - \text{desc}_u(c'_i)) - 2(k - \text{desc}_u(c''_i)) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

We have that

$$\ell(u, w) = p_i - 2(b - \text{desc}_u(c_i)) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

and

$$\ell(u, wt_{kb}) \geq p'_i + p''_i - 2(b - \text{desc}_u(c'_i)) - 2(k - \text{desc}_u(c''_i)) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

If c'_i has a descent, then c''_i does not, and we are done. If c''_i has a descent, then

$$w(k) = u(a_{i,j-1}) < u(b) < u(a_{i,p_i}) = w(b)$$

by Lemma 4.4, so

$$\ell(w) < \ell(wt_{bk}) \leq \ell(w) + 2(b - k) - 1$$

so

$$\ell(u, wt_{kb}) = \ell(wt_{bk}) - \ell(u) \leq p'_i + p''_i - 2(b - \text{desc}_u(c'_i)) - 2(b - \text{desc}_u(c''_i)) + 2(b - k) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

Since we already have the opposite inequality, we are done if c''_i has a descent, since $wt_{bk} \triangleleft w$. If neither has a descent but c_i does, then this descent must be at k . We have that $wt_{bk}(k) = u(a_{i,p_i}) > u(b)$ by Lemma

4.4, and $wt_{bk}(b) = u(a_{i,j-1}) < u(b)$. Furthermore, $u(k) > u(b) > u(r) > u(a_{i,j-1})$ for all r with $k < r < b$ since k is a descent. Thus $wt_{bk} \triangleleft w$.

$a_{i,q} < r < b$, so the length decreases by $2(b - a_{i,q}) - 1$ as desired. If $u(a_{i,j+1}) > u(k)$

To show that the q' we have chosen is unique, let $r > k$ be any index other than q' such that $\ell(ut_{kr}) = \ell(u) + 1$. Suppose first that r is not in any of the cycles c_i . Recall that c_1 is the cycle that contains k . Then $t_{kr}c_1$ is a cycle in the disjoint cycle decomposition of $(ut_{kr})^{-1}w$ containing both k and r , which are both larger than $k - 1$. This violates the condition of Lemma 4.2 that in order for $ut_{kr} \xrightarrow{k-1} w$ we must have that each cycle contains exactly one element larger than $k - 1$, hence we have $ut_{kr} \xrightarrow{k-1} w$. If r is in one of the cycles, since $r > k$ and $r \neq q'$ this cycle cannot be c_1 . Say the cycle containing r is $c_2 = (a'_{p_2}, \dots, a'_1, r)$. Then

$$\begin{aligned} t_{kr}c_1c_2 &= t_{kr}(r, a'_{p_2}, \dots, a'_1)(a_{p_1}, \dots, a_1, q') \\ &= (r, a'_{p_2}, \dots, a'_1, k)(k, a_{j-1}, \dots, q', a_{p_1}, \dots, a_{j+1}) \\ &= (r, a'_{p_2}, \dots, a'_1, k, a_{j-1}, \dots, q', a_{p_1}, \dots, a_{j+1}) \end{aligned}$$

is a single cycle disjoint from all the others containing both k and r , which are both greater than $k - 1$, hence we have $ut_{kr} \xrightarrow{k-1} w$ by Lemma 4.2. The uniqueness of q' follows.

To prove the nonexistence of q , consider the same cycle decomposition $c_1, \dots, c_n$. Let $q < k$ be such that $\ell(ut_{qk}) = \ell(u) + 1$. If c_1 (the cycle that contains k) does not contain q , then $t_{qk}c_1$ is a cycle containing both k and some integer larger than k . If some other cycle c_2 contains q , $t_{qk}c_1c_2$ is disjoint from all the other cycles and contains both k and some integer larger than k , similarly to the calculation above. Thus if c_1 does not contain q then the disjoint cycle decomposition of $(ut_{qk})^{-1}w$ contains the cycle $t_{qk}c_1$ or $t_{qk}c_1c_2$, which implies that $ut_{qk} \xrightarrow{k-1} w$ by Lemma 4.2 since the cycle contains two elements larger than $k - 1$. To handle the case where c_1 contains q , in that case t_{qk} splits c_1 into two disjoint cycles as follows. Assume $c_1 = (a_{p_1}, \dots, q')$ with $a_j = k$ and $a_{j'} = q$. We must have that $j' > j$ since $u(q) < u(k)$. Then

$$\begin{aligned} t_{qk}c_1 &= t_{qk}(a_{p_1}, \dots, q, \dots, k, \dots, q') \\ &= t_{qk}(q, \dots, k, \dots, q', a_{p_1}, \dots, a_{j'+1}) \\ &= t_{qk}(q, a_{j'-1}, \dots, k)(k, \dots, q', a_{p_1}, \dots, a_{j'+1}) \\ &= (q, a_{j'-1}, \dots, a_{j+1})(a_{p_1}, \dots, a_{j'+1}, k, \dots, q') \end{aligned}$$

The cycle $(a_{p_1}, \dots, k, \dots, q')$ contains two elements larger than $k - 1$, implying that $ut_{qk} \xrightarrow{k-1} w$ by Lemma 4.2, as desired. $\square$

We organize the inductive step of Theorem 2.1 around the position k , distinguishing whether or not $u(k) = w(k)$. The following lemma records the two combinatorial facts about the interaction of $\xrightarrow{k}$ and $\xrightarrow{k-1}$ at position k that make this dichotomy exhaustive.

Lemma 4.6. *Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer.*

- (i) *If $u(k) = w(k)$, then $u \xrightarrow{k} w$ if and only if $u \xrightarrow{k-1} w$.*
- (ii) *If $u(k) \neq w(k)$ and $u \xrightarrow{k} w$, then $u \xrightarrow{k-1} w$.*

Proof. Both statements follow from the cycle description of Lemma 4.2, according to which $u \xrightarrow{m} w$ if and only if in the disjoint cycle decomposition of $u^{-1}w$ each cycle contains exactly one index exceeding m , subject to the length conditions. If $u(k) = w(k)$ then k is a fixed point of $u^{-1}w$, hence lies in no nontrivial cycle; the constraint distinguishing level k from level $k - 1$ concerns only the indices exceeding $k - 1$ that occur in the cycles, and k is not among them. Thus the cycle conditions for level k and level $k - 1$ coincide, proving (i). For (ii), if $u(k) \neq w(k)$ then k lies in a nontrivial cycle c of $u^{-1}w$; if $u \xrightarrow{k} w$, the level- k condition forces k to be the unique index of c exceeding $k - 1$ that is admissible at level k , but then c contains an index $b > k$ as its cycle top, so c contains two indices $\geq k$, violating the level- $(k - 1)$ condition. Hence $u \xrightarrow{k-1} w$. $\square$

The heart of the induction is the following cancellation, which replaces the informal “the sums are empty” step of the classical argument. When $u(k) = w(k)$ the diagonal and straight recursion terms already reproduce $c_k(p; u, w)$ through the single-variable recurrence for factorial elementary polynomials (Lemma 4.11); the reflection sums and the quantum term must therefore cancel among themselves.

Lemma 4.7 (Reflection-drop cancellation). *Suppose $u, w \in S_\infty$, let $k > 1$, and suppose $u(k) = w(k)$. Fix $1 \leq p \leq k$. Then*

$$\sum_{a < k, wt_{ak} \triangleleft w} \text{sign}(a-k) \kappa_{ak} c_{k-1}(p-1; u, wt_{ak}) + \sum_{b > k, wt_{kb} \triangleleft w} \text{sign}(b-k) \kappa_{kb} c_{k-1}(p-1; u, wt_{kb}) + q_{k-1} c_{k-2}(p-2; u, w) = 0,$$

where $\kappa_{rs} = q_{rs}$ if $wt_{rs} \triangleleft w$ is length-decreasing and $\kappa_{rs} = 1$ if it is length-increasing. In fact the two reflection sums contribute at most one nonzero summand in total: it occurs precisely when $u \xrightarrow{k-2} w$ (equivalently, when the quantum term is nonzero), it is indexed by some $a < k$, and it equals $-q_{k-1} c_{k-2}(p-2; u, w)$.

Proof. Throughout we use that $u(k) = w(k)$, so that k is a fixed point of $\pi := u^{-1}w$.

Step 1: no reflection t_{kb} with $b > k$ contributes. Suppose $b > k$ and $wt_{kb} \triangleleft w$ with $c_{k-1}(p-1; u, wt_{kb}) \neq 0$, so $u \xrightarrow{k-1} v$ for $v := wt_{kb}$. Then $u^{-1}v = \pi t_{kb}$; since k is a fixed point of π and $b > k$, right multiplication by t_{kb} adjoins k to the π -cycle containing b (or creates the 2-cycle (kb)). Either way the resulting cycle of $u^{-1}v$ contains two indices exceeding $k-1$, namely k and an index $\geq b > k$, contradicting the level- $(k-1)$ condition of Lemma 4.2. Hence only reflections t_{ak} with $a < k$ can contribute, and each such summand carries the sign $\text{sign}(a-k) = -1$.

Step 2: the reflection sum has at most one nonzero term, occurring precisely when $u \xrightarrow{k-2} w$. A reflection t_{ak} with $a < k$ contributes precisely when $u \xrightarrow{k-1} v$ for $v := wt_{ak}$; here $u^{-1}v = \pi t_{ak}$, so t_{ak} adjoins the fixed point k to the π -cycle containing a (or creates the 2-cycle (ak) when a is fixed by π). By Lemma 4.2, $u \xrightarrow{k-1} v$ requires this enlarged cycle to contain exactly one index exceeding $k-1$; as it already contains k , it may contain no further index $\geq k$. Since k is fixed by π , an index occurring in a nontrivial π -cycle and exceeding $k-1$ is in fact $\geq k+1$, unless it equals $k-1$. Thus the enlarged cycle is admissible only if the π -cycle receiving k has all indices $\leq k-1$, i.e. its top equals $k-1$; the π -cycles being disjoint, at most one $a < k$ contributes. Such a configuration exists exactly when $u \xrightarrow{k-2} w$: if $u \xrightarrow{k-2} w$ and $k-1$ is not fixed by π , then the π -cycle through $k-1$ has top $k-1$ (any second index $\geq k-1$ would be $\geq k+1$, violating the level- $(k-2)$ condition), while if $k-1$ is fixed the choice $a = k-1$ creates the 2-cycle $(k-1 k)$; conversely a contributing t_{ak} forces a π -cycle of top $k-1$ (or $k-1$ fixed with $a = k-1$), whence $u \xrightarrow{k-2} w$. This is exactly the condition under which the quantum term is nonzero.

Step 3: the contributing term cancels the quantum term. Assume $u \xrightarrow{k-2} w$, let t_{a^*k} ($a^* < k$) be the unique contributing reflection, and set $v := wt_{a^*k}$. By Step 2 the cycle of $u^{-1}v$ containing k has top k . Since v agrees with w off $\{a^*, k\}$ and $v(a^*) = w(k) = u(k)$, both a^* and $k-1$ leave the agreement set in passing to level $k-1$, and comparison of the invariants of Definition 2.1 gives

$$P_{k-1}(u, v) = P_{k-2}(u, w), \quad n_{k-1}(u, v) = n_{k-2}(u, w) + 1$$

(using $n_m = m - |P_m|$). Hence $c_{k-1}(p-1; u, v)$ and $c_{k-2}(p-2; u, w)$ carry the *same* factorial elementary polynomial $E_{(p-2)-n_{k-2}(u, w), |P|}(y_P; z)$ with $P = P_{k-2}(u, w)$. Finally the q -weights match. Write $\kappa_{a^*k} = 1$ if t_{a^*k} is length-increasing and $\kappa_{a^*k} = q_{a^*k}$ if length-decreasing. As the q -weight is multiplicative over the cycles of $u^{-1}(\cdot)$ and only the cycle receiving k is altered (from top $k-1$ to top k), the definition of the weights yields

$$\kappa_{a^*k} q_{k-1}(u, v) = q_{k-1} q_{k-2}(u, w).$$

Therefore the contributing summand is

$$\text{sign}(a^* - k) \kappa_{a^*k} c_{k-1}(p-1; u, v) = -\kappa_{a^*k} q_{k-1}(u, v) E_P = -q_{k-1} q_{k-2}(u, w) E_P = -q_{k-1} c_{k-2}(p-2; u, w),$$

cancelling the quantum term. When $u \not\xrightarrow{k-2} w$ the quantum term vanishes and, by Step 2, no reflection contributes; all three terms are 0. In every case the displayed sum is 0. $\square$

Remark 4.8. The content of Lemma 4.7 is the identification of the quantum term $q_{k-1} c_{k-2}(p-2; u, w)$ with the unique level- $(k-1)$ neighbour v satisfying $u \xrightarrow{k-1} v$, $u \not\xrightarrow{k-2} v$, $P_{k-1}(u, v) = P_{k-2}(u, w)$ and $n_{k-1}(u, v) = n_{k-2}(u, w) + 1$, obtained by adjoining the fixed point k to the π -cycle of top $k-1$. The

connecting reflection t_{a^*k} may be length-increasing or length-decreasing; in either case its weight κ_{a^*k} satisfies the balance $\kappa_{a^*k}q_{k-1}(u, w) = q_{k-1}q_{k-2}(u, w)$. Uniqueness of v is a consequence of the cycle structure (there is a single π -cycle of top $k-1$); this intrinsic matching by the invariants (P, n) is what the $u(k) \neq w(k)$ branch of Lemma 4.10 invokes.

When $u(k) \neq w(k)$ the position k lies in a nontrivial cycle of $\pi = u^{-1}w$, and the reflection sums of the vanishing identity below no longer telescope against the diagonal; instead they cancel among themselves and against the quantum term, one q -weight at a time. The following lemma isolates this cancellation. It is the moved-case counterpart of Lemma 4.7, and its non-quantum instances specialize at $M = 1$ to the fixed-point-free reflection cancellation of the classical Pieri rule [1].

Lemma 4.9 (Moved reflection cancellation). *Let $u, w \in S_\infty$, let $k > 1$, and suppose $u \not\stackrel{k}{\rightarrow} w$ with $u(k) \neq w(k)$. Let $\pi = u^{-1}w$, let C be the π -cycle containing k — nontrivial, since $u(k) \neq w(k)$ — and set $B = \max(C)$. Fix $1 \leq p \leq k$ and a q -monomial $q^M \neq q_{k-1}(u, w)$, and let $\mathcal{G}_M$ be the set of Monk summands of (4.3)–(4.4) of q -weight q^M , augmented by the quantum term $q_{k-1}c_{k-2}(p-2; u, w)$ of (4.5) when $q^M = q_{k-1}q_{k-2}(u, w)$. Then the summands of $\mathcal{G}_M$ sum to 0.*

Proof. A neighbour $v = wt$, with $t \in \{t_{ak} : a < k\} \cup \{t_{kb} : b > k\}$, contributes to $\mathcal{G}_M$ only if $u \xrightarrow{k-1} v$; writing $\sigma = u^{-1}v = \pi t$, Lemma 4.2 requires every cycle of σ to contain exactly one index exceeding $k-1$. Since t moves π only at its two positions, one of which is $k \in C$, the passage $\pi \rightsquigarrow \sigma$ either *splits* C (when the other position lies in C) or *merges* C with the component of the other position (when it does not); every other π -cycle is unchanged and so must already satisfy the level- $(k-1)$ condition.

Up-neighbours are unique and force C to have exactly the indices k, B above $k-1$. Let $v = wt_{kb}$ with $b > k$. If $b \notin C$, the merge produces a cycle containing k, b and $B \geq b$, hence two indices exceeding $k-1$: inadmissible. Thus $b \in C$, and t_{kb} splits C into a piece through k and a piece through b ; admissibility forces each to hold a single index above $k-1$, so b is that index for its piece. As $B = \max(C) > k-1$, this yields $b = B$ and shows C has exactly the two indices k, B above $k-1$, with $B > k$. Hence a contributing up-neighbour, if present, is unique and equals wt_{kB} .

Down-neighbours. Let $v = wt_{ak}$ with $a < k$. If $a \in C$ (a *split*), admissibility assigns the indices of C above $k-1$ one to each nontrivial piece: if C has three or more such indices no split is admissible; if it has exactly two, k and $B > k$, the split separates them (and then wt_{kB} is the up-neighbour); if it has only k , so $B = k$, the split peels a fixed point off C . If $a \notin C$ (a *merge*), then $C \cup (\text{component of } a)$ must retain a single index above $k-1$, namely k , forcing $B = k$ and the component of a to have all indices $\leq k-1$. We record the resulting q -weights.

(1) *The case $B = k$.* Either a is adjoined as a fixed point — then $v(a) = w(k) = u(a)$ reinstates below k the agreement lost at k , the cycle tops are unchanged, and the summand carries weight $q_{k-1}(u, w)$ — or the component absorbed into C is a nontrivial cycle, whose unique index above $k-2$ is then $k-1$ (being $\leq k-1$ and $> k-2$); the merged cycle acquires the extra factor q_{k-1} , and a direct comparison of the invariants of Definition 2.1 gives

$$P_{k-1}(u, v) = P_{k-2}(u, w), \quad n_{k-1}(u, v) = n_{k-2}(u, w) + 1, \quad \kappa_{ak}q_{k-1}(u, v) = q_{k-1}q_{k-2}(u, w),$$

the very identities of Lemma 4.7, with the π -cycle of top $k-1$ now attached to C in place of the fixed point k ; the computation is otherwise verbatim, and by uniqueness of the π -cycle of top $k-1$ there is at most one such v .

(2) *The case $B > k$.* Here only the split with $C = \{k, B\}$ above $k-1$ occurs, and it separates k from B .

Cancellation. Suppose first $q^M = q_{k-1}q_{k-2}(u, w)$, so the quantum term lies in $\mathcal{G}_M$ and $u \xrightarrow{k-2} w$; then every π -cycle has exactly one index above $k-2$, in particular $B = k$, and by the up-paragraph $\mathcal{G}_M$ contains no up-neighbour. By (1) the summands of weight q^M reduce to the single merge down-neighbour v with $P_{k-1}(u, v) = P_{k-2}(u, w)$; then $c_{k-1}(p-1; u, v)$ and $c_{k-2}(p-2; u, w)$ carry the same factorial elementary polynomial $E_{(p-2)-n_{k-2}(u, w), (k-2)-n_{k-2}(u, w)}(y_{P_{k-2}(u, w)}; z)$, and the down summand, of sign $\text{sign}(a-k) = -1$, equals $-q_{k-1}c_{k-2}(p-2; u, w)$ and cancels the quantum term. (If no such v exists the quantum term already vanishes.)

Now suppose $q^M \neq q_{k-1}(u, w)$, $q_{k-1}q_{k-2}(u, w)$. By (1) every merge or peel down carries weight $q_{k-1}(u, w)$ or $q_{k-1}q_{k-2}(u, w)$, hence lies outside $\mathcal{G}_M$; so every summand of $\mathcal{G}_M$ is of split type with $C = \{k, B\}$, $B > k$ (whence also $u \xrightarrow{k-1} w$). For each such summand $P_{k-1}(u, v) = P_{k-1}(u, w)$: the up wt_{kB} agrees with w below

k , while a split down wt_{ak} agrees with w below k except at $a \in C$, where it takes the value $w(k)$; but a is not fixed by π , and $u(a) \neq w(k)$ (otherwise, by (1), the summand would have weight $q_{k-1}(u, w)$), so no agreement below k is created or destroyed. Consequently all summands of $\mathcal{G}_M$ share the factorial elementary factor $E_0 = E_{(p-1)-n, (k-1)-n}(y_P; z)$, where $P = P_{k-1}(u, w)$ and $n = n_{k-1}(u, w)$. Finally, when $C = \{k, B\}$ there are exactly two admissible neighbours through position k : the up wt_{kB} , and — since C has a single descent (Lemma 4.3) and, as in the proof of Lemma 4.5, exactly one covering split of C separates k from B — a single down wt_{a^*k} . The two reflections differ by the 3-cycle (a^*, B, k) ; each resolves the cycle C into two cycles of tops k and B while leaving the other cycles untouched, so by the multiplicativity of the q -weight (Definition 2.1) and Lemma 4.4, which fixes $\text{desc}_u(C)$ and hence the drop factors, both $\kappa_{kB}q_{k-1}(u, wt_{kB})$ and $\kappa_{a^*k}q_{k-1}(u, wt_{a^*k})$ equal the weight q^M of $\mathcal{G}_M$. With opposite signs $\text{sign}(B - k) = +1$ and $\text{sign}(a^* - k) = -1$ and common coefficient E_0 , the up and the down cancel. Hence $\mathcal{G}_M = 0$. $\square$

The remaining case of the induction is when $u \not\stackrel{k}{\rightarrow} w$, where the target coefficient is 0.

Lemma 4.10 (Vanishing). *Suppose $u, w \in S_\infty$, let $k > 1$, and suppose $u \not\stackrel{k}{\rightarrow} w$. Then for all $1 \leq p \leq k$,*

$$(y_{w(k)} - z_{k-p+1})c_{k-1}(p-1; u, w) + \sum_{b > k, wt_{kb} \triangleleft w} \kappa_{kb}c_{k-1}(p-1; u, wt_{kb}) - \sum_{a < k, wt_{ak} \triangleleft w} \kappa_{ak}c_{k-1}(p-1; u, wt_{ak}) \\ + c_{k-1}(p; u, w) + q_{k-1}c_{k-2}(p-2; u, w) = 0,$$

where $\kappa_{rs} = q_{rs}$ if $wt_{rs} \triangleleft w$ is length-decreasing and $\kappa_{rs} = 1$ if it is length-increasing.

Proof. Suppose first that $u(k) = w(k)$. Then by Lemma 4.6(i) also $u \stackrel{k-1}{\rightarrow} w$, so both $c_{k-1}(p-1; u, w)$ and $c_{k-1}(p; u, w)$ vanish; hence the diagonal term and the straight recursion term are 0, and the remaining reflection sums and the quantum term cancel by Lemma 4.7. This settles the case $u(k) = w(k)$ completely.

Now suppose $u(k) \neq w(k)$. Since $u \not\stackrel{k}{\rightarrow} w$, Lemma 4.2 shows the cycle of $u^{-1}w$ containing k has a top $b > k$, so $c_k(p; u, w) = 0$; writing F for the right-hand side (4.2)–(4.5), we must show $F = 0$.

We grade by the cohomological degree $\deg q_i = 2$. Under this grading (4.1) is homogeneous, and the q_i are algebraically independent over $\mathbb{Z}[y, z]$. By the definition of c_m , each summand of F is a monomial in the q_i times a factorial elementary polynomial in $\mathbb{Z}[y, z]$; collecting by q -monomial we write $F = \sum_M q^M F_M$ with $F_M \in \mathbb{Z}[y, z]$, and

$$F = 0 \iff F_M = 0 \text{ for every } q\text{-monomial } M.$$

Fix M and let $\mathcal{G}_M$ be the collection of summands of F whose q -weight equals q^M ; here the diagonal (4.2) and the straight term $c_{k-1}(p; u, w)$ carry weight $q_{k-1}(u, w)$, the quantum term carries $q_{k-1}q_{k-2}(u, w)$, and the Monk summand of (4.3)–(4.4) reaching $w' = wt_{pp,k}$ carries $\kappa q_{k-1}(u, w')$ with $\kappa \in \{1, q_{ak}, q_{kb}\}$ its drop-weight. We show $F_M = 0$ in two exhaustive cases.

Case (i): $\mathcal{G}_M$ contains the diagonal or the straight term. These carry weight $q_{k-1}(u, w)$, so $q^M = q_{k-1}(u, w)$; in particular $u \stackrel{k-1}{\rightarrow} w$. Write $P = P_{k-1}(u, w)$, $n = n_{k-1}(u, w)$, $m = k-1-n = |P|$ and $d = p-1-n$. Set $a_0 = u^{-1}(w(k))$; since $u(k) \neq w(k)$ and $w(k)$ is a value of u occurring at a position $< k$ (this holds whenever the diagonal is nonzero), we have $a_0 < k$. The Monk summand indexed by t_{a_0k} then satisfies $u \stackrel{k-1}{\rightarrow} wt_{a_0k}$ with

$$P_{k-1}(u, wt_{a_0k}) = P \cup \{w(k)\}, \quad n_{k-1}(u, wt_{a_0k}) = n-1,$$

carries weight $q_{k-1}(u, w)$, and is the *only* Monk summand in $\mathcal{G}_M$. Dividing $\mathcal{G}_M$ by the common factor $q_{k-1}(u, w)$, its three contributions read

$$(y_{w(k)} - z_{k-p+1})E_{d,m}(y_P; z) + E_{d+1,m}(y_P; z) - E_{d+1,m+1}(y_{P \cup \{w(k)\}}; z),$$

which vanishes by the single-variable recurrence of Lemma 4.11 adjoining the variable $y_{w(k)}$ (the z -index matches because $m-d = k-p$). Hence $F_M = 0$.

Case (ii): $\mathcal{G}_M$ contains neither the diagonal nor the straight term. Then $q^M \neq q_{k-1}(u, w)$, and $\mathcal{G}_M$ consists of Monk summands of (4.3)–(4.4) together with the quantum term when $q^M = q_{k-1}q_{k-2}(u, w)$. By Lemma 4.9 these summands sum to 0, so $F_M = 0$.

As the two cases are exhaustive, every $F_M = 0$, so $F = 0$. $\square$

We now record the recurrence satisfied by the quantum factorial elementary polynomials, which drives the induction.

Lemma 4.11. *For all integers p and k we have*

$$E_{p,k}^q(x; z) = (x_k - z_{k-p+1})E_{p-1,k-1}^q(x; z) + E_{p,k-1}^q(x; z) + q_{k-1}E_{p-2,k-2}^q(x; z),$$

with the conventions $E_{0,k}^q = 1$ for $k \geq 0$ and $E_{p,k}^q = 0$ if $p < 0$ or $p > k$. Moreover, the ordinary factorial elementary polynomials satisfy the single-variable recurrence

$$E_{a,m}(y_1, \dots, y_m; z) = E_{a,m-1}(y_1, \dots, y_{m-1}; z) + (y_m - z_{m-a+1})E_{a-1,m-1}(y_1, \dots, y_{m-1}; z).$$

The first recurrence is the quantization of the classical relation $E_{p,k} = (x_k - z_{k-p+1})E_{p-1,k-1} + E_{p,k-1}$; the extra term $q_{k-1}E_{p-2,k-2}$ is the only feature absent from the non-quantum setting. The classical Pieri formula, obtained by discarding all quantum terms, is proved in [1] by induction on k using the first recurrence in the case $q = 0$; we adapt that argument, the essential new point being the bookkeeping of the powers of q .

Proof of Theorem 2.1. For a permutation w , write $[\mathfrak{S}_w^q]F$ for the coefficient of $\mathfrak{S}_w^q(x; y)$ in an expansion F of a quantum double Schubert polynomial in the basis $\{\mathfrak{S}_w^q(x; y)\}$. With this notation the theorem is the assertion that, for all u and all $0 \leq p \leq k$,

$$[\mathfrak{S}_w^q](\mathfrak{S}_u^q(x; y)E_{p,k}^q(x; z)) = \begin{cases} q_k(u, w) E_{p-n_k(u,w), k-n_k(u,w)}(y_{P_k(u,w)}; z) & \text{if } u \xrightarrow{k} w, \\ 0 & \text{otherwise.} \end{cases}$$

For brevity, when $u \xrightarrow{m} w$ we set

$$c_m(a; u, w) = q_m(u, w) E_{a-n_m(u,w), m-n_m(u,w)}(y_{P_m(u,w)}; z),$$

and $c_m(a; u, w) = 0$ if $u \not\xrightarrow{m} w$; note that $c_m(a; u, w) = 0$ whenever $a < n_m(u, w)$ or $a > m$, since then the factorial elementary factor vanishes. The claim is that $[\mathfrak{S}_w^q](\mathfrak{S}_u^q E_{p,k}^q) = c_k(p; u, w)$.

We argue by strong induction on k , the statement for level k being understood to include all p simultaneously.

Base case $k = 1$. Only $p \in \{0, 1\}$ give a nonzero $E_{p,1}^q$. For $p = 0$ we have $E_{0,1}^q = 1$ and the claim is trivial, since $u \xrightarrow{1} w$ with $q_1 = 1$, $n_1 = 0$ only for $w = u$. For $p = 1$ we have $E_{1,1}^q(x; z) = x_1 - z_1$, and Lemma 4.1 with $i = j = 1$ gives

$$(x_1 - z_1) \mathfrak{S}_u^q(x; y) = (y_{u(1)} - z_1) \mathfrak{S}_u^q(x; y) + \sum_{\substack{b > 1 \\ \ell(ut_{1b}) = \ell(u) + 1}} \mathfrak{S}_{ut_{1b}}^q(x; y) + \sum_{\substack{b > 1 \\ \ell(ut_{1b}) = \ell(u) - 2(b-1) + 1}} q_{1b} \mathfrak{S}_{ut_{1b}}^q(x; y).$$

The permutations w with $u \xrightarrow{1} w$ are exactly $w = u$ and $w = ut_{1b}$ for $b > 1$ with $u \triangleleft ut_{1b}$, since Definition 2.1 permits at most one reflection when $k = 1$. For $w = u$ we have $q_1 = 1$, $n_1 = 0$, $P_1(u, u) = \{u(1)\}$, and $E_{1,1}(y_{u(1)}; z) = y_{u(1)} - z_1$, matching the diagonal term. For $w = ut_{1b}$ we have $n_1 = 1$, $P_1 = \emptyset$, and $E_{0,0}(\cdot; z) = 1$; if the step is length-increasing then $q_1(u, w) = 1$, matching the second sum, and if it is length-decreasing then $q_1(u, w) = q_{1b}$, matching the third sum. All other coefficients vanish on both sides. This proves the base case.

Inductive step. Fix $k > 1$ and $1 \leq p \leq k$, and assume the claim at all levels $m < k$. By Lemma 4.11,

$$(4.1) \quad \mathfrak{S}_u^q E_{p,k}^q = (x_k - z_{k-p+1})(\mathfrak{S}_u^q E_{p-1,k-1}^q) + \mathfrak{S}_u^q E_{p,k-1}^q + q_{k-1} \mathfrak{S}_u^q E_{p-2,k-2}^q.$$

By the inductive hypothesis, $\mathfrak{S}_u^q E_{p-1,k-1}^q = \sum_v c_{k-1}(p-1; u, v) \mathfrak{S}_v^q$. Applying Lemma 4.1 with $i = k$, $j = k - p + 1$ to each $\mathfrak{S}_v^q$ and extracting the coefficient of $\mathfrak{S}_w^q$ from (4.1), we obtain

$$(4.2) \quad [\mathfrak{S}_w^q](\mathfrak{S}_u^q E_{p,k}^q) = (y_{w(k)} - z_{k-p+1}) c_{k-1}(p-1; u, w)$$

$$(4.3) \quad + \sum_{\substack{b > k, \, wt_{kb} \triangleleft w \\ \ell(w) = \ell(wt_{kb}) + 1}} c_{k-1}(p-1; u, wt_{kb}) + \sum_{\substack{b > k, \, wt_{kb} \triangleleft w \\ \ell(w) = \ell(wt_{kb}) - 2(b-k) + 1}} q_{kb} c_{k-1}(p-1; u, wt_{kb})$$

$$(4.4) \quad - \sum_{\substack{a < k, \, wt_{ak} \triangleleft w \\ \ell(w) = \ell(wt_{ak}) + 1}} c_{k-1}(p-1; u, wt_{ak}) - \sum_{\substack{a < k, \, wt_{ak} \triangleleft w \\ \ell(w) = \ell(wt_{ak}) - 2(k-a) + 1}} q_{ak} c_{k-1}(p-1; u, wt_{ak})$$

$$(4.5) \quad + c_{k-1}(p; u, w) + q_{k-1} c_{k-2}(p-2; u, w),$$

where we used that $\text{sign}(b - k) = 1$ for $b > k$ and $\text{sign}(a - k) = -1$ for $a < k$, and that a summand in (4.3)–(4.4) is present precisely when $u \xrightarrow{k-1} wt_{kb}$, respectively $u \xrightarrow{k-1} wt_{ak}$ (otherwise $c_{k-1} = 0$). We must show this equals $c_k(p; u, w)$. We distinguish three cases according to whether $u(k) = w(k)$ and, when $u(k) \neq w(k)$, whether $u \xrightarrow{k} w$; by Lemma 4.6 these cases are exhaustive and, in particular, the combination $u(k) \neq w(k)$ with both $u \xrightarrow{k} w$ and $u \xrightarrow{k-1} w$ never occurs.

Case 1: $u(k) = w(k)$. By Lemma 4.6(i) we have $u \xrightarrow{k} w$ if and only if $u \xrightarrow{k-1} w$; suppose first that both hold. Since $u(k) = w(k)$,

$$n_k(u, w) = n_{k-1}(u, w), \quad P_k(u, w) = P_{k-1}(u, w) \cup \{u(k)\}, \quad q_k(u, w) = q_{k-1}(u, w),$$

the last equality holding because no reflection with first index k occurs. Writing $m = k - n_k(u, w) = |P_k(u, w)|$ and $a = p - n_k(u, w)$, the diagonal contribution (4.2) and the straight recursion term $c_{k-1}(p; u, w)$ of (4.5) give

$$(y_{u(k)} - z_{k-p+1}) c_{k-1}(p-1; u, w) + c_{k-1}(p; u, w) = q_{k-1}(u, w) \left[(y_{u(k)} - z_{m-a+1}) E_{a-1, m-1}(y_{P_{k-1}}; z) + E_{a, m-1}(y_{P_{k-1}}; z) \right],$$

using $k - p + 1 = m - a + 1$. By the single-variable recurrence of Lemma 4.11 with the new variable $y_{u(k)}$, the bracket equals $E_{a, m}(y_{P_k}; z)$, so these two contributions sum to $q_k(u, w) E_{a, m}(y_{P_k}; z) = c_k(p; u, w)$. The remaining contributions are the reflection sums (4.3), (4.4) and the quantum term $q_{k-1} c_{k-2}(p-2; u, w)$ of (4.5), which sum to 0 by Lemma 4.7. Hence the total is $c_k(p; u, w)$, as required. If instead $u(k) = w(k)$ but $u \not\xrightarrow{k} w$ (equivalently $u \xrightarrow{k-1} w$), then $c_k(p; u, w) = 0$ and the diagonal and straight terms both vanish, while the reflection sums and quantum term again cancel by Lemma 4.7; the total is $0 = c_k(p; u, w)$.

Case 2: $u \xrightarrow{k} w$ but $u \not\xrightarrow{k-1} w$. By Lemma 4.5 there is a unique $b > k$ with $wt_{kb} \triangleleft w$ and $u \xrightarrow{k-1} wt_{kb}$; set $v = wt_{kb}$. This is exactly the unique nonvanishing summand of (4.3), and by Lemma 4.5 the sums (4.4) vanish, as does the diagonal (4.2) and the terms (4.5) (since $u \not\xrightarrow{k-1} w$ and $u \not\xrightarrow{k-2} w$). Moreover $P_k(u, w) = P_{k-1}(u, v)$ and $n_k(u, w) = n_{k-1}(u, v) + 1$, so

$$E_{p-n_k(u, w), k-n_k(u, w)}(y_{P_k}; z) = E_{(p-1)-n_{k-1}(u, v), (k-1)-n_{k-1}(u, v)}(y_{P_{k-1}(u, v)}; z).$$

If $\ell(w) = \ell(v) + 1$ then the reflection t_{kb} is length-increasing, so by Definition 2.1 $q_k(u, w) = q_{k-1}(u, v)$ and the contribution is $c_{k-1}(p-1; u, v) = c_k(p; u, w)$. If instead $\ell(w) = \ell(v) - 2(b-k) + 1$ then t_{kb} is length-decreasing, Definition 2.1 gives $q_k(u, w) = q_{k-1}(u, v) q_{kb}$, and the contribution is $q_{kb} c_{k-1}(p-1; u, v) = c_k(p; u, w)$. In both cases the coefficient equals $c_k(p; u, w)$, as required. This is the step at which the quantum drop terms of Lemma 4.1 and the recursive definition of $q_k(u, w)$ interlock.

Case 3: $u \not\xrightarrow{k} w$ (with $u(k) \neq w(k)$). Here $c_k(p; u, w) = 0$, so we must show the coefficient (4.2)–(4.5) vanishes. This is exactly the assertion of Lemma 4.10. (The subcase $u(k) = w(k)$ with $u \not\xrightarrow{k} w$ was already treated at the end of Case 1.)

In all cases $[\mathfrak{S}_w^q](\mathfrak{S}_u^q E_{p, k}^q) = c_k(p; u, w)$, completing the induction and the proof. $\square$

REFERENCES

- [1] SAMUEL, M. J. A Molev-Sagan type formula for double Schubert polynomials. *J. Pure Appl. Algebra* 228, 7 (2024), 107636.
- [2] SOTTILE, F. Pieri's rule for flag manifolds and Schubert polynomials. *Ann. Inst. Fourier* 46, 1 (1996), 89–110.